

Yiming Huang

IMPERIAL · DEPARTMENT OF COMPUTING

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Research

My research interests primarily focus on developing novel topological deep learning methods for understanding complex and higher-order graph structures. I am also interested in generative models, graph representation learning, foundation models, higher-order network analysis, AI4Science, and their interdisciplinary applications.

- Topological Deep Learning (TDL)
- AI4Science
- Generative Models
- Graph Representation Learning
- Network Science
- Foundation Models

Education

Imperial College London

PH.D. IN COMPUTER SCIENCE

London, UK

Sep. 2024 - Present

- Supervised by [Prof. Tolga Birdal](#)
- Awarded the prestigious Lee Family Scholarship (granted to two recipients annually across Imperial)

UESTC (University of Electronic Science and Technology of China)

M.SC. IN COMPUTER SCIENCE

Chengdu, China

Sep. 2021 - Jun. 2024

- Supervised by [Prof. Linyuan Lü](#)
- GPA: **3.8** / 4.0
- Joint training programs with the University of Science and Technology of China (USTC) and the Yangtze Delta Research Institute (Huzhou)

NUIST (Nanjing University of Information Science and Technology)

B.S. IN COMPUTER SCIENCE

Nanjing, China

Sep. 2017 - Jun. 2021

- GPA: **4.1** / 5.0 (**3.9** / 4.0), top **2** in major

Publications

(*: Equal contribution. [C] = Conference (2), [J] = Journal (6), [W] = Workshop (2).)

[W1] Higher-order grammar representations for molecular topology generation and learning

[Yiming Huang](#), Yujie Zeng, Vijay Prakash Dwivedi, Simone Foti, Jianming Wang, Jure Leskovec, Tolga Birdal

ICLR 2026 Workshop on Foundation Models for Science: Real-World Impact and Science-First Design. [\[Paper\]](#)

TLDR: We introduce HGR, a higher-order grammar representation that converts combinatorial complexes into sequences, enabling direct higher-order topology generation and molecular foundation modeling, together with RingDiv, a ring-enriched benchmark.

Key words: Topological Deep Learning, Generative Model, Molecular Foundation Model

[C1] HOG-Diff: Higher-Order Guided Diffusion for Graph Generation

[Yiming Huang](#), Tolga Birdal

International Conference on Learning Representations (ICLR), 2026 [\[Paper\]](#) [\[Website\]](#)

TLDR: We introduce HOG-Diff, a coarse-to-fine graph generation framework that explicitly exploits higher-order topological cues.

Key words: Topological Deep Learning, Generative Model

[W2] Cellular-Guided Graph Generative Model

[Yiming Huang](#), Tolga Birdal

ICLR 2025 Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy, 2025 [\[Paper\]](#)

Key words: Topological Deep Learning, Generative Model

[C2] Higher-Order Graph Convolutional Network with Flower-Petals Laplacians on Simplicial Complexes

[Yiming Huang*](#), Yujie Zeng*, Qiang Wu, Linyuan Lü

Proceedings of the AAAI conference on artificial intelligence (AAAI), 2024 [\[Paper\]](#) [\[Appendix\]](#) [\[Poster\]](#)

TLDR: Propose the Flower-Petals (FP) representation and HiGCN, a higher-order GNN with provably stronger expressiveness.

Key words: Topological Deep Learning, Graph Representation Learning, Network Science

[J1] Identifying key players in complex networks via network entanglement

[Yiming Huang](#), Hao Wang, Xiao-Long Ren, Linyuan Lü

Communications Physics p. 19. 2024 [\[Paper\]](#)

Key words: Network Science, Influence Maximization

[J2] Cooperative Network Learning for Large-Scale and Decentralized Graphs

Qiang Wu*, [Yiming Huang*](#), Yujie Zeng, Linyuan Lü

IEEE Transactions on Computational Social Systems. 2026 [\[Paper\]](#)

Key words: Graph Representation Learning, Network Science

[J3] Influential simplices mining via simplicial convolutional networks

Yujie Zeng*, Yiming Huang*, Qiang Wu, Linyuan Lü

Information Processing & Management p. 103813. 2024 [Paper]

Key words: Topological Deep Learning, Graph Representation Learning, Network Science

[J4] Identifying vital nodes through augmented random walks on higher-order networks

Yujie Zeng*, Yiming Huang*, Xiao-Long Ren, Linyuan Lü

Information Sciences p. 121067. 2024 [Paper]

Key words: Network Science, Influence Maximization

[J5] A novel coherence-based quantum steganalysis protocol

Zhiguo Qu, Yiming Huang, Min Zheng

Quantum Information Processing p. 362. 2020 [Paper]

Key words: Quantum Computing

[J6] An efficient logic-guided attack strategy on logical-physical coupled networks

Mengying Geng, Ling Zhang, Min Hui, Yiming Huang, Feng Qing, Mengqiao Xu, Changjun Fan, Guangquan Cheng

Chaos, Solitons and Fractals 202. p. 117396. 2026 [Paper]

Projects

Topology-aware RNA Dynamic Learning

Imperial, UK

2026 - Present

- Topology-aware JEPA learning flexible RNA representations from single conformers, with multi-conformer and MD as soft dynamics signal.
- Higher-order supervision via cell complexes and contact occupancy to capture loops, junctions, and conformer shifts in dynamic RNA.

Topological Generative Models for Molecules

Imperial, UK

Sep. 2024 - Present

- Designing topology-aware diffusion frameworks and grammar representations for molecular graph generation.
- Built *RingDiv*, a ring-enriched benchmark exposing limitations of current molecular generators.
- Advancing scalable molecular/graph foundation models guided by higher-order structural cues.
- Selected: [C1] HOG-Diff (ICLR 2026), [W1] HGR (ICLR 2026 Workshop), [W2] CG3 (ICLR 2025 Workshop).

Higher-order Graph Representation Learning and Network Science

UESTC & USTC, China

Sep. 2021 - Jun. 2024

- Developed *Flower-Petals* Laplacians and higher-order GNNs with provably stronger expressiveness.
- Applied higher-order topological methods to influence maximization, node ranking, and decentralized graph learning.
- Contributed to the Chinese monograph *Graph Machine Learning*; collaborated with Prof. Jure Leskovec.
- Selected: [C2] HiGCN (AAAI 2024), [J1] Network Entanglement (Comm. Phys. 2024), [J3] ISMnet (IPM 2024), [J4] HoRW (Inf. Sci. 2024).

Honors & Awards

Total: 1 international award, more than 10 provincial and above awards; 7 patents; 1 national research project.

AWARDS

- 2023 **Gold Prize**, “Internet+” College Students Innovation and Entrepreneurship Competition (**Top 1%**)
- 2021 **Honor Graduate**, (Top 10%)
- 2020 **Finalist**, Mathematical Contest In Modeling (MCM) (**Top 1%**)
- 2020 **Project Leader**, National Training Program of Innovation and Entrepreneurship for Undergraduates

NUIST

SCHOLARSHIPS

- 2024-2028 **Lee Family Scholarship**, (**Top 2**, two recipients annually across Imperial)
- 2021-2023 **First-Class**, Academic Scholarship of Master Degree student (Top 10%, received each academic year)
- 2017-2020 **First-Class**, Academic Scholarship of Bachelor Degree student (Top 10%, received each academic year)

Imperial

UESTC

NUIST

Academic Services

Conference Reviewing

NeurIPS (2025), ICML (2025, 2026), AAAI (2025), ICLR (2025, 2026), ECCV (2026)

Journal Reviewing

Nature Communications, Information Sciences, Neurocomputing, Entropy, Applied Sciences, Journal of Computational Science, IEEE Transactions on Artificial Intelligence (TAI), IEEE Transactions on Computational Social Systems (TCSS), Physica A: Statistical Mechanics and its Applications, International Journal of Modern Physics C

Organizing Committee

LOGML (London Geometry and Machine Learning) Summer School 2026 [Website]